

Detecting Epileptic Seizures in Mesoscale GCaMP Imaging Data

Preprocessing pipelines and detection algorithms for cortex-wide widefield calcium imaging — classical signal processing compared with machine learning

A literature review of peer-reviewed and openly documented methods
Scope: widefield (mesoscale) cortex-wide imaging • detection and classification algorithms •
preprocessing and signal-extraction pipelines

Compiled 25 August 2026

Index

1. Scope, Definitions and How to Read This Report	4
1.1 What this report covers	4
1.2 Key terms used throughout	4
1.3 The state of the field, in one paragraph	4
2. What a Seizure Looks Like in Mesoscale GCaMP Data	4
2.1 Amplitude and duration signatures	4
2.2 Spatial and propagation signatures	5
2.3 Spectral and cell-type separation	5
2.4 The pre-ictal signature (relevant to prediction, not just detection)	5
3. Acquisition Choices that Constrain Detection Performance	6
3.1 Sampling and optical access	6
3.2 Indicator and mouse line — a first-order confound	6
3.3 Ground truth acquired at the same time	6
4. Preprocessing and Signal-Extraction Pipeline	6
4.1 Step 1 — Frame-level corrections	6
4.2 Step 2 — $\Delta F/F$ and baseline estimation	7
4.3 Step 3 — Hemodynamic correction	7
4.4 Step 4 — Denoising, compression and dimensionality reduction	7
4.5 Step 5 — Spatial parcellation and atlas registration	8
4.6 Step 6 — Confound handling before detection	8
4.7 Available open-source pipelines	8
5. Classical Signal-Processing Detection Algorithms	8
5.1 Global amplitude and spatial-extent thresholding (the published standard)	8
5.2 Peak-prominence event detection with duration gating (IEDs)	9
5.3 Time-domain features borrowed from EEG detectors	9
5.4 Time-frequency and band separation	9
5.5 Spatiotemporal and wave-based characterisation	9
5.6 Template matching for pre-ictal detection	10
5.7 Strengths and limitations of the classical family	10
6. Machine Learning and Deep Learning Approaches	10
6.1 Unsupervised structure discovery (currently the most established ML use)	10
6.2 Supervised classical machine learning	11
6.3 Deep learning	11

6.4 The data problem, and how to work around it	11
6.5 Strengths and limitations of the learned family	11
7. Side-by-Side Comparison	12
8. Validation, Ground Truth and Reporting	12
8.1 Ground truth	12
8.2 Metrics that are appropriate here	12
8.3 Statistical practice worth copying	13
9. A Recommended Pipeline	13
9.1 Stage A — Preprocessing (fixed, validated once)	13
9.2 Stage B — High-sensitivity candidate generation (classical)	13
9.3 Stage C — Classification and rejection (learned)	13
9.4 Stage D — Characterisation and readout	13
10. Open Challenges and Research Gaps	14
11. References	14

1. Scope, Definitions and How to Read This Report

1.1 What this report covers

- **Preparation:** widefield (mesoscale, single-photon) imaging of genetically encoded calcium indicators of the GCaMP family across the dorsal cortex of rodents, typically head-fixed and awake, with a field of view spanning most of one or both hemispheres.
- **Problem:** automatically identifying ictal events (seizures) and, closely related, interictal epileptiform discharges (IEDs) in the resulting image time series — including where a seizure starts, when it starts, and how it spreads.
- **Two halves of the problem, treated separately:** (i) the preprocessing / signal-extraction pipeline that turns raw frames into an analysable fluorescence signal, and (ii) the detection and classification algorithms that operate on that signal.
- **Two analytical families, compared side by side:** classical signal processing (thresholds, feature engineering, wave analysis) and machine learning / deep learning.

1.2 Key terms used throughout

- $\Delta F/F$ — fractional change in fluorescence relative to a per-pixel baseline F_0 ; the standard unit of mesoscale calcium signal.
- **Ictal / interictal** — during a seizure / between seizures. **IED** (interictal epileptiform discharge, or interictal spike) is a brief, high-amplitude pathological event outside seizures.
- **Emergence zone** — the set of pixels above seizure threshold in the first frame of a seizure; **recruitment** is the subsequent entry of other pixels into the seizure, often summarised as a rank map.
- **Hemodynamic contamination** — distortion of the GCaMP signal by blood volume and oxygenation changes, because haemoglobin absorbs light at GCaMP excitation and emission wavelengths.
- **CCF** — the Allen Mouse Brain Common Coordinate Framework, the reference atlas into which cortical images are usually registered [18].

1.3 The state of the field, in one paragraph

Mesoscale calcium imaging of seizures is an established but small literature, and it is dominated by *classical* detection: amplitude and spatial-extent thresholds, peak prominence criteria, and hand-built spatiotemporal descriptors. Machine learning appears in this literature mostly in supporting roles — unsupervised decomposition, artifact classification, atlas registration — rather than as the seizure detector itself. Deep learning, meanwhile, is the dominant paradigm for seizure detection in EEG and intracranial EEG, where large labelled corpora exist. The most useful way to read this report is therefore as a description of a mature classical toolkit (Section 5) alongside a largely untapped but well-signposted machine-learning opportunity (Section 6).

Citation convention. Numbers in square brackets refer to the numbered reference list in Section 11. Every substantive methodological claim is tied to a source there.

2. What a Seizure Looks Like in Mesoscale GCaMP Data

Detector design should follow from signal phenomenology. The published characterisations converge on a small set of features that separate epileptic activity from physiological cortical activity at the mesoscale.

2.1 Amplitude and duration signatures

- Both prolonged ictal events and brief interictal events are **orders of magnitude larger** than normal sensory-evoked responses in the same preparation, which is the single most exploitable property for detection [3].
- In a genetic mouse model (a *Kcnt1* gain-of-function knock-in crossed to Snap25-GCaMP6s), IEDs were separable from normal events as a joint **high-intensity, short-duration** population: peak prominence above the 99th percentile of wild-type event prominence, with a mean width-at-half-peak below roughly 0.9 s [1].
- Seizures in the same dataset were mostly brief (median around 4.5 s) and strongly **clustered in trains** (median inter-seizure interval around 14 s), so detectors must handle events that recur within seconds of each other [1].
- Important caveat: in that study the IED population did not form a discrete cluster but a **continuous extension** of the normal event distribution, implying that any IED / non-IED boundary is a modelling choice rather than a natural division [1].

2.2 Spatial and propagation signatures

- Seizures emerge from a **consistent, restricted set of cortical areas** rather than uniformly across cortex, and the areas recruited earliest are the same ones that host emergence on other occasions [1].
- After emergence the seizing area grows along a roughly **sigmoidal curve** to maximum extents on the order of 9–24 mm², with growth rate itself informative (seizures following soon after a previous seizure grow faster) [1].
- In an acute focal model in visual cortex, seizures and interictal events both begin as **standing waves** — synchronous elevation in the focus and in homotopic locations of higher areas — then propagate both locally and to distal homotopic regions, respecting the connectivity that supports normal sensory processing [3].
- Propagation is not always contiguous: seizures make **long-range jumps** to non-adjacent tissue, and those jumps occur between strongly synaptically connected areas far more often than chance [1, 4].
- Two distinct wave types coexist — a slow, outward-moving ictal wavefront and fast, inward-moving ictal discharges — separated by roughly two orders of magnitude in speed [5]. A detector that assumes a single propagation velocity will mis-describe both.

2.3 Spectral and cell-type separation

- Wide-field calcium recording has been used as an **optical electrocorticogram**: separating high-frequency (neuronal) from low-frequency (glial) components showed that neuronal waves propagate rapidly but stay largely confined to the focus, while glial waves are triggered at ictal onset and spread slowly far beyond it [6].
- Practical consequence: band-splitting the $\Delta F/F$ trace is not merely denoising — different bands carry different biology, and the slow band can extend a detected event well past the electrographic seizure.

2.4 The pre-ictal signature (relevant to prediction, not just detection)

- Spatial *patterns* immediately before seizures were not abnormal — matching patterns occur interictally and even in wild-type animals — but their **temporal persistence** was: activity became progressively concentrated in the impending emergence zone, significantly so from about 9.4 s before onset, and before peak amplitude exceeded interictal levels [1].
- This means an amplitude-only detector discards a usable pre-ictal signal; the informative quantity is the **persistence of a spatial template**, not its height.

3. Acquisition Choices that Constrain Detection Performance

Several decisions made at the rig determine what any downstream algorithm can achieve. They are listed here because they are frequently the true limiting factor on detector sensitivity.

3.1 Sampling and optical access

- Typical acquisition for seizure work is around **20 Hz**, with alternate frames given over to a reflectance channel for hemodynamic correction — which halves the effective calcium sampling rate and directly limits how finely onset timing and wave speed can be resolved [1].
- Optical access is either a **glass cranial window** or an **intact skull** rendered semi-transparent with cyanoacrylate; the latter is chronic-friendly and cheap but adds scattering. Scattering can bias spatial detection thresholds, and a published control is to **rescale per-pixel thresholds by F_0** and verify that recruitment maps are unchanged [1, 19].
- Cortex-wide protocols and their acquisition parameters are documented in step-by-step form [11] and in an epilepsy-specific protocol chapter [2, 25].

3.2 Indicator and mouse line — a first-order confound

- **This is the most important single caveat in the field.** Multiple laboratories have reported spontaneous, interictal-spike-like electrical events with accompanying fluorescence transients in certain GCaMP6 transgenic genotypes, most prominently *Emx1-Cre;Camk2a-tTA;Ai93*. The events occur above 0.1 Hz, exceed 10% $\Delta F/F$ in widefield signals, and cover large cortical regions [7].
- A well-tuned seizure or IED detector run on such a line will faithfully report epileptiform events that are a property of the **reporter, not the disease model**. Line choice must be reported and, where possible, controlled.
- The same report notes that these events were often **missed when only a limited cortical region was imaged**, which is itself an argument for cortex-wide fields of view [7].
- Lines used in published seizure imaging include Snap25-2A-GCaMP6s (pan-neuronal) [1, 24] and Thy1-GCaMP6 lines; red-shifted indicators (jRGECO1a) are an option where haemoglobin absorption is especially problematic.

3.3 Ground truth acquired at the same time

- **Simultaneous LFP or EEG** is the electrographic reference standard and is used in essentially every serious mesoscale seizure study [3, 4, 6].
- **Body video** supplies behavioural semiology (for example Straub tail, back arching, convulsions) used to screen candidate events [1].
- **Locomotion** (treadmill or wheel encoder) must be recorded: running drives large, widespread cortical activity that is a major false-positive source, and published analyses explicitly restrict interictal event statistics to non-running periods [1].

4. Preprocessing and Signal-Extraction Pipeline

The pipeline below reflects the consensus order of operations in published mesoscale work. Deviations matter: several steps interact badly with the very events being detected.

4.1 Step 1 — Frame-level corrections

- **Dark-frame subtraction** from all illuminated frames, then **de-interleaving** of the fluorescence and reflectance channels [1].

- **Motion correction** by rigid registration; even in head-fixed animals the brain moves within the cranium, and preparation-related contrast changes masquerade as signal [16]. Motion correction is the first stage of the standard open pipelines [12, 13].
- **Exclusion of pathological non-seizure events before decomposition:** where seizures transition into cortical spreading depression, the CSD epochs were removed prior to matrix decomposition because they distorted the resulting components [1].

4.2 Step 2 — $\Delta F/F$ and baseline estimation

- Baseline choice is not neutral in epileptic data. Because large events are frequent, a simple temporal mean is contaminated; a robust published alternative is to take each pixel's **average over its lowest 20% of raw values** as F_0 [1].
- Baseline drift across a long session is handled by **linear detrending** plus a very low cut-off high-pass filter (for example second-order Butterworth at 0.01 Hz) [1].
- Recommendation: compute and store the baseline map, and check it is not itself modulated by seizure burden across a session.

4.3 Step 3 — Hemodynamic correction

GCaMP excitation and emission wavelengths sit inside the haemoglobin absorption spectrum, so blood volume and oxygenation changes contaminate the measured fluorescence [10, 27]. Options, in increasing order of rigour:

- **Frequency-domain suppression.** Low-pass filtering or a band-stop notch over the heartbeat band (for example a second-order Butterworth stop-band between 7 and 14 Hz) removes the strongest cardiac artifact [1, 10].
- **Second-wavelength reflectance subtraction.** Alternate-frame illumination (commonly a 530 nm green LED, or violet/isosbestic excitation) yields a reflectance image; its fractional change is subtracted from the GCaMP $\Delta F/F$ [1, 9, 10].
- **Regression and Beer-Lambert models.** Multi-wavelength backscatter can be related to haemoglobin absorption through a modified Beer-Lambert relationship; accuracy then hinges on mean scattering path lengths that are hard to measure [8, 9].
- **Spatially detailed pixelwise regression (recommended default).** A linear form of the Beer-Lambert relationship fitted independently at every pixel, avoiding the estimation of physical parameters. In a direct comparison across three Cre-driver lines this *Spatial Model* outperformed the other commonly used methods, particularly in regions where separation is otherwise difficult [8].
- **Open caveat specific to epilepsy.** These corrections are validated on physiological activity. Seizures produce very large, non-linear neurovascular changes, so correction assumptions may fail exactly during the events of interest. Control experiments with activity-independent fluorophores (GFP mice) are the standard way to bound this error [8].

4.4 Step 4 — Denoising, compression and dimensionality reduction

- **SVD / PCA.** Singular value decomposition of the $\Delta F/F$ stack serves as denoising and compression in one step; a rank-50 reconstruction was used for all downstream analyses in the *Kcnt1* study, with the right singular vectors detrended and filtered before reconstruction [1]. Note that PCA/SVD force orthogonality, which is not physiologically motivated [20].
- **Penalised matrix decomposition (PMD)** is used as the compression stage in the *wfield* pipeline, combined with motion correction and hemodynamic correction [12, 14].
- **Localised NMF (LocaNMF)** produces spatially localised, non-negative components that are easier to map onto anatomy than SVD modes [12].

- **Warning for seizure work:** a global basis learned from a whole session is dominated by the highest-variance events. Rare, enormous events (seizures, CSD) can either be absorbed into a few components or distort the basis for everything else. Either exclude them before fitting, or verify that the reconstruction preserves ictal waveforms.

4.5 Step 5 — Spatial parcellation and atlas registration

- **Allen CCF registration** is the field standard for reporting where seizures emerge. A published three-step scheme combines a within-animal affine transform on histogram-equalised session-average images, a cross-animal alignment based on functional topography from **seed-pixel correlation maps**, and a refinement pass against the group-average map [1, 18].
- **MesoNet** automates this: a deep-learning model locates nine cortical landmarks from a single raw fluorescence image, and a fully convolutional network delimits brain boundaries, producing Allen-atlas region segmentation with performance comparable to human raters [15].
- **Data-driven functional segmentation.** Independent component analysis on densely sampled widefield recordings recovers on the order of 115 functional domains per hemisphere; a **random forest classifier** trained on spatial and temporal component metrics then separates neural from artifact components at human-level performance [16].
- **Cheap alternative.** A regular grid of ROIs (for example 50×50 covering the field of view) is sufficient as a first-pass detection substrate, with full-resolution reconstruction only around detected candidates [1].

4.6 Step 6 — Confound handling before detection

- Exclude or model **locomotion epochs**; retain the encoder trace as a regressor and as an audit trail for false positives [1, 23].
- Interpolate over **surface vasculature** gaps and smooth before any morphological operation on spatial maps (a published recipe uses region filling followed by a Gaussian kernel with standard deviation of about 4 pixels) [1].
- Decide explicitly on **global signal regression**. Some open toolboxes regress the global signal by default [21]; for seizure detection this is usually the wrong choice, since generalised events are largely global and would be partly removed.

4.7 Available open-source pipelines

- **wfield** (Python) — motion correction, penalised matrix decomposition, hemodynamic correction and LocaNMF, with an atlas overlay viewer [12].
- **WidefieldImager** (MATLAB) — acquisition software plus localised SVD dimensionality reduction and hemodynamic correction [13].
- **NeuroCAAS** — cloud execution of the above widefield pipeline as a reproducible, scalable job chain [14].
- **MesoNet** (Python) — registration and segmentation, GUI and CLI [15].
- **FLOW portraits** — optical-flow-based spatiotemporal visualisation, described in Section 5.5 [17].

5. Classical Signal-Processing Detection Algorithms

Everything in this section is in active published use on mesoscale calcium data, or is directly transplantable from electrophysiological seizure detection.

5.1 Global amplitude and spatial-extent thresholding (the published standard)

The most completely documented mesoscale seizure detector combines an amplitude criterion with a rate-of-change criterion, then a spatial criterion for boundaries [1]:

- Compute two per-frame vectors: (i) the **fraction of pixels above 20% $\Delta F/F$** , and (ii) the **mean pixelwise $\Delta F/F$ difference between successive frames**.
- Smooth both with a 10 s moving average; detect **coincident peaks** (within 15 s) in the envelope of the first vector and in the second.
- Screen candidates against body video and cortical $\Delta F/F$ for behavioural correlates.
- Set a **per-animal seizure threshold** at the 99th percentile of peak amplitudes of non-seizure, non-IED interictal events.
- Define event boundaries by pixel counts: seizure start is the first frame with at least 1% of pixels above threshold; end of the growth phase is the frame where the above-threshold pixel count is maximal; termination is the first trailing frame below 1%.
- Finally, remove candidates that meet IED criteria, since IEDs near seizures also cross threshold.

Why it works: the two vectors are near-orthogonal descriptors — one captures *how much* cortex is active, the other *how fast* it is changing — and requiring coincidence suppresses both slow global drifts and isolated fast artifacts.

5.2 Peak-prominence event detection with duration gating (IEDs)

- Two-stage detection: run peak detection on grid-ROI mean traces with a **prominence threshold** (7.5% was used), merge peaks in different ROIs occurring within 250 ms into a single event, then repeat detection at full pixel resolution within each candidate [1].
- Classify as an IED when the event is **brief** (mean width at half peak below about 0.9 s) and of **unusually high peak intensity** (above the 99th percentile of wild-type prominence) [1].
- Measure intensity from a map in which each pixel carries its peak prominence, smoothed with a 2-D Gaussian, taking the highest 2-D peak; measure duration from the co-located value in the corresponding width-at-half-max map [1].
- The equivalent electrophysiological approach — a scatter plot of prominence against width of all LFP peaks, looking for a discrete high-prominence, short-duration cluster — is a useful cross-check that the optical events are real [7].

5.3 Time-domain features borrowed from EEG detectors

- **Line length, signal energy, amplitude and slope** computed in sliding windows are the workhorses of real-time electrographic seizure detection and map directly onto ROI $\Delta F/F$ traces; a recent closed-loop detection chip combines exactly these four with an adaptive threshold rule [29].
- Adapt window lengths to the calcium time constant, not to EEG conventions: at 10–20 Hz effective sampling with GCaMP6s kinetics, sub-second windows are noise-dominated.
- Per-animal, per-session **normalisation** (z-scoring against interictal statistics) is essential because absolute $\Delta F/F$ depends on expression level, window quality and scattering.

5.4 Time-frequency and band separation

- Split the trace into a **high-frequency neuronal band** and a **low-frequency band** that carries glial and slow components; the two dissociate during seizure evolution, so detecting on the combined signal blurs onset and grossly overestimates spatial extent [6].
- Wavelet or spectrogram representations also enable image-based classifiers later (Section 6.3), since a time-frequency map is a natural input to a convolutional network [30, 31].

5.5 Spatiotemporal and wave-based characterisation

- **Recruitment rank maps.** For each pixel, record the time it first crosses seizure threshold and rank it; averaging rank maps across seizures reveals the stereotyped order in which areas are recruited [1].

- **Emergence-zone heat maps.** Accumulate first-frame above-threshold pixel sets across seizures to localise susceptible territory, then express as fractions of each atlas area [1].
- **Long-range jump detection.** Apply an extended-minima transform to smoothed, vasculature-filled recruitment rank maps to find secondary foci that are non-contiguous with the primary one [1].
- **Optical flow and finite-time Lyapunov exponents.** The FLOW framework converts the image series into time-varying vector fields by optic flow, then extracts ridges of the FTLE field to produce compact portraits showing regions of initiation and termination and the direction and extent of spread — capturing coherent structures that modal decomposition represents poorly [17].
- **Standing-wave versus travelling-wave tests.** Comparing activity at the focus with activity at homotopically matched distal locations distinguishes synchronous onset from contiguous radial spread [3].

5.6 Template matching for pre-ictal detection

- Build a binary template from the **top 5% of pixel values** in a frame (after vasculature filling and Gaussian smoothing) [1].
- Score any other frame by **overlap divided by the size of the smaller mask**, giving a 0–1 match score [1].
- Construct the null distribution from templates of ordinary interictal events of the same type, so that the comparison is against equally similar activity rather than against chance [1].
- The detectable pre-ictal quantity is a **ramping increase in match score**, significant from roughly 9.4 s before onset in the reference study [1].

5.7 Strengths and limitations of the classical family

- **Strengths:** interpretable and auditable; needs no labelled training set; works with the small event counts typical of these experiments (a flagship study identified 52 seizures in 34.5 h of imaging) [1]; parameters map onto physical quantities that reviewers can check.
- **Limitations:** thresholds require per-animal calibration; performance is brittle to changes in window quality, expression level and frame rate; low sensitivity to focal or low-amplitude events that never reach a global criterion; weak intrinsic rejection of movement artifacts, which must be handled by exclusion rather than by the detector; and spatial *pattern* information is used descriptively after detection rather than as a detection feature.

6. Machine Learning and Deep Learning Approaches

6.1 Unsupervised structure discovery (currently the most established ML use)

- **Hierarchical clustering of events by spatial profile.** Represent each detected event as a vector of mean peak $\Delta F/F$ per atlas area, then cluster on average correlation distance, choosing cluster number by **silhouette analysis**. In the reference study this yielded nine IED clusters and twenty non-IED event clusters, and made it possible to compare like-with-like across genotypes [1].
- **ICA plus random forest artifact rejection.** Independent component analysis separates neural sources from vascular, hemodynamic and motion sources; a random forest trained on component-level spatial and temporal metrics then labels components automatically at human performance [16].
- **SVD across animals.** Applying SVD to an event-type $\times$ animal intensity matrix yields a shared pattern of variation and a per-animal **scaling factor**; that scalar predicted epileptic activity burden in mutant animals [1]. This is a useful template for deriving low-dimensional, animal-level biomarkers.

6.2 Supervised classical machine learning

- **Random forests, SVMs and k-NN on engineered features** are well validated for early seizure detection in intracranial EEG and transfer naturally to ROI-wise calcium features (line length, energy, spectral power, spatial extent, growth rate).
- **Graph-based feature construction.** A visibility-graph transformation of widefield GCaMP6f time series has been used to decode cortical behavioural states, with an explicit comparison across feature sets and classifiers — a directly reusable feature-engineering recipe for a seizure classifier [28].
- These methods are the pragmatic middle ground: they exploit labelled data where it exists, but remain trainable on the tens-to-hundreds of events realistically available.

6.3 Deep learning

- **Precedent on this exact data type.** An end-to-end CNN, and a CNN combined with a recurrent network, classified behavioural state directly from cortex-wide calcium imaging frames with high accuracy and remained accurate with time windows as short as roughly 0.17–0.5 s [32]. This establishes that raw mesoscale frames are a viable network input and that short windows suffice.
- **Segmentation networks** (U-Net) are already used on this data for anatomical delineation [15]; the same architecture family applies to segmenting the seizing territory frame by frame, which would replace hand-set spatial thresholds.
- **Architectures transferable from EEG.** The electrographic literature has converged on 1-D and 2-D CNNs, CNN-LSTM and bidirectional LSTM hybrids, convolutional autoencoders feeding a classifier, multi-spectral feature learning, and more recently transformer and graph-attention models that treat channels as graph nodes [30, 31, 33, 34, 35]. A cortex-wide image is a natural graph (atlas areas as nodes), which makes graph-attention models an unusually good fit.
- **Few-shot personalisation is realistic.** In intracranial recordings a CNN reached near-maximal ictal-pattern detection accuracy with only about 10–30 seed samples per patient [36] — encouraging for per-animal fine-tuning on a handful of electrographically confirmed seizures.
- **The gap:** as of this review no widely adopted, validated deep-learning seizure detector specific to widefield calcium imaging has been established; the deep-learning seizure literature remains overwhelmingly EEG-based [33, 34].

6.4 The data problem, and how to work around it

- **Label scarcity and extreme class imbalance.** Seizures occupy a vanishing fraction of recording time, and a well-powered study may contain only tens of events across a handful of animals [1].
- **Leakage risk.** With few animals, random frame-wise splits will place frames from the same seizure in training and test. **Leave-one-animal-out** and leave-one-session-out are the only defensible protocols.
- **Weak supervision.** Use simultaneous LFP/EEG to auto-label optical data, converting an expensive manual task into an automatic one [3, 4, 6].
- **Anomaly-detection framing.** Train on interictal data only and flag deviations; this sidesteps the imbalance problem entirely and matches the fact that seizures are, by construction, rare extremes.
- **Augmentation and simulation.** Spatial flips, rotations and elastic deformations are physiologically defensible after atlas registration; biophysical seizure models can additionally supply synthetic training data [5].

6.5 Strengths and limitations of the learned family

- **Strengths:** uses the full spatiotemporal pattern rather than summary scalars; learns artifact rejection instead of requiring exclusion rules; adapts across animals with modest fine-tuning; supports short decision windows suitable for real-time use.

- **Limitations:** demands labelled events that this field does not yet have at scale; risks learning rig- or animal-specific idiosyncrasies; harder to defend in a methods section; and offers no interpretable onset definition unless paired with an explicit segmentation output.

7. Side-by-Side Comparison

Criterion	Classical signal processing	Machine learning / deep learning
Labelled data required	None; thresholds calibrated from the animal's own interictal distribution [1].	Tens to hundreds of confirmed events; weak labels from simultaneous LFP make this tractable [3, 4].
Interpretability	High — every parameter is a physical quantity (% $\Delta F/F$, seconds, mm^2).	Low by default; recoverable with saliency or occlusion analysis, as used on cortex-wide calcium decoders [32].
Use of spatial pattern	Descriptive: pattern analysed after detection (rank maps, emergence zones) [1].	Native: the spatial pattern is an input feature, which is where the untapped gain lies.
Artifact robustness	Weak; movement and hemodynamic artifacts handled by exclusion rules [1].	Learnable; ICA plus random forest already does this at human level for components [16].
Transfer across animals / rigs	Poor; per-animal recalibration is standard practice [1].	Moderate to good; few-shot personalisation is effective in the electrographic analogue [36].
Real-time / closed loop	Proven in electrophysiology with line-length and energy features on fixed hardware [29].	Feasible — sub-second windows already give accurate cortex-wide classification [32] — but unproven for seizures.
Pre-ictal detection	Demonstrated via template-match persistence, roughly 9 s of lead time [1].	Plausible and likely superior, but not yet demonstrated on mesoscale calcium data.
Current evidence base in widefield calcium	Strong; this is what published seizure imaging studies actually use [1, 3, 4, 6].	Thin for detection; strong for preprocessing, segmentation and registration [15, 16].

Bottom line. The two families are complements rather than rivals. Classical methods give a defensible, low-data detector today; learned methods are the route to artifact robustness, cross-animal transfer and pre-ictal prediction, and the labelled data needed to build them can be generated automatically by running a classical detector against simultaneous electrophysiology.

8. Validation, Ground Truth and Reporting

8.1 Ground truth

- **Primary:** simultaneous LFP or EEG, scored electrographically and independently of the optical data [3, 4, 6].
- **Secondary:** behavioural semiology from synchronised body video [1].
- **Negative controls:** wild-type littermates for event-rate baselines [1], and activity-independent fluorophore (GFP) animals to bound hemodynamic correction error [8].
- **Reporter control:** confirm the mouse line is not one associated with spontaneous epileptiform events [7].

8.2 Metrics that are appropriate here

- Report **event-wise sensitivity** together with **false detections per hour**, not frame-wise accuracy — with seizures occupying a fraction of a percent of recording time, frame accuracy is uninformative.

- Under this imbalance, **area under the precision-recall curve** is the appropriate summary statistic; it is the metric of choice in the comparable electrographic literature [36].
- Report **onset latency error** in seconds against the electrographic reference; for closed-loop ambitions this matters more than classification accuracy.
- For spatial claims, report **emergence-zone overlap** and recruitment-rank correlation against the electrographic or histological reference, and verify that maps are stable across window type (glass versus thinned skull) [1].

8.3 Statistical practice worth copying

- **Permutation and shuffle tests** rather than parametric tests for group differences in event rate, duration and intensity, with correction for multiple comparisons across event types [1].
- **Purpose-built null distributions**: when testing whether a pre-ictal pattern is special, compare it against the most similar interictal events, not against random frames [1].
- **Session weighting** when animals contribute unequal numbers of sessions [1].

9. A Recommended Pipeline

A staged design that uses each family where it is strongest, and that is buildable with the data volumes typical of this field.

9.1 Stage A — Preprocessing (fixed, validated once)

- Dark-frame subtraction → channel de-interleaving → rigid motion correction [12].
- $\Delta F/F$ with a bottom-20th-percentile per-pixel baseline [1].
- Hemodynamic correction by spatially detailed pixelwise regression [8], followed by a heartbeat band-stop and a 0.01 Hz high-pass [1].
- SVD or PMD compression with CSD epochs excluded from the fit; retain a moderate rank and verify ictal waveform fidelity in the reconstruction [1, 12].
- Register to the Allen CCF, automatically via MesoNet with manual spot-checks [15, 18].

9.2 Stage B — High-sensitivity candidate generation (classical)

- Run the two-vector amplitude-plus-rate-of-change detector with a **deliberately permissive** threshold, accepting a high false-positive rate [1].
- In parallel, run prominence-based event detection on a 50×50 ROI grid to catch IEDs and focal events the global detector misses [1].
- Tag every candidate with locomotion state and a raw movement-energy score.

9.3 Stage C — Classification and rejection (learned)

- Train a classifier on candidates labelled automatically by simultaneous LFP, using leave-one-animal-out validation.
- Start with a random forest on engineered features (peak amplitude, prominence, width at half max, spatial extent, growth rate, atlas-area profile, band ratio); escalate to a CNN-recurrent model on short frame windows only if the feature model plateaus [32].
- Treat the ICA-plus-random-forest artifact classifier as a reusable component rather than rebuilding it [16].

9.4 Stage D — Characterisation and readout

- Emergence-zone maps, recruitment rank maps, growth curves and long-range jump detection [1].
- FLOW portraits for wave initiation, termination and direction of spread [17].

- Template-match persistence traces in the 20 s preceding each detected onset, for pre-ictal analysis [1].
- Per-animal burden summary via the event-type SVD scaling factor [1].

10. Open Challenges and Research Gaps

- **Hemodynamic correction during ictal events is unvalidated.** Existing models are fitted and tested on physiological activity; the neurovascular changes accompanying seizures are far outside that range [6, 8]. A GFP-control study specifically during seizures would be a publishable contribution in itself.
- **There is no shared benchmark dataset or annotation standard** for mesoscale seizure imaging, which is the proximate reason deep learning has not arrived here. The EEG field's progress tracked its public corpora closely [33].
- **Reporter-line confounds remain incompletely characterised** beyond the lines already flagged [7], and detectors are rarely reported alongside a line-level control.
- **Real-time optical closed-loop detection is essentially unexplored.** The components exist — sub-second window classification [32] and mature threshold-based real-time detection hardware [29] — but have not been combined on widefield calcium.
- **The IED / non-IED boundary may not be a real categorical distinction** [1], which argues for continuous severity scores rather than binary labels in future detectors.
- **Generalisation across seizure models is untested:** acute pharmacological models (4-aminopyridine, bicuculline) [4, 6] and genetic models [1, 24] differ in onset dynamics, and a detector tuned on one should not be assumed to transfer.
- **Depth ambiguity.** Widefield signal originates from a mixture of superficial and deeper sources [27], so an apparent emergence zone is a surface projection of a three-dimensional event; combining mesoscale detection with targeted depth recording remains an open design problem [22, 26].

11. References

All sources were consulted online. Where an author list could not be verified from the accessed record, the entry is listed by title.

- [1] Tobin, W.F. & Weston, M.C. (2023). Distinct features of interictal activity predict seizure localization and burden in a mouse model of childhood epilepsy. *Journal of Neuroscience*, 43(27), 5076–5091. doi:10.1523/JNEUROSCI.2205-22.2023 — <https://www.jneurosci.org/content/43/27/5076>
- [2] Tobin, W.F. & Weston, M.C. (2025). Mesoscale Ca^{++} imaging of seizures and interictal activity in models of chronic epilepsy. In: Niemeyer, J.E., Ma, H. & Schwartz, T.H. (eds), *Awake Behaving Mesoscopic Brain Imaging*, Neuromethods vol. 214, pp. 233–254. Humana, New York. doi:10.1007/978-1-0716-4120-0_10
- [3] Rossi, L.F., Wykes, R.C., Kullmann, D.M. & Carandini, M. (2017). Focal cortical seizures start as standing waves and propagate respecting homotopic connectivity. *Nature Communications*, 8(1), 217. doi:10.1038/s41467-017-00159-6 — <https://www.nature.com/articles/s41467-017-00159-6>
- [4] Liou, J.Y., Ma, H., Wenzel, M., Zhao, M., Baird-Daniel, E., Smith, E.H., Daniel, A., Emerson, R., Yuste, R., Schwartz, T.H. & Schevon, C.A. (2018). Role of inhibitory control in modulating focal seizure spread. *Brain*, 141(7), 2083–2097. doi:10.1093/brain/awy116
- [5] Liou, J.Y. et al. (2020). A model for focal seizure onset, propagation, evolution, and progression. *eLife*, 9, e50927. doi:10.7554/eLife.50927 — <https://elifesciences.org/articles/50927>
- [6] Daniel, A.G.S., Laffont, P., Zhao, M., Ma, H. & Schwartz, T.H. (2015). Optical electrocorticogram (OECOG) using wide-field calcium imaging reveals the divergence of neuronal and glial activity during acute rodent seizures. *Epilepsy & Behavior*, 49, 61–65. doi:10.1016/j.yebeh.2015.04.036
- [7] Steinmetz, N.A., Buettner, C., Lecoq, J., ... Carandini, M. & Harris, K.D. (2017). Aberrant cortical activity in multiple GCaMP6-expressing transgenic mouse lines. *eNeuro*, 4(5), ENEURO.0207-17.2017.

- [8] Valley, M.T., Moore, M.G., Zhuang, J., Mesa, N., Castelli, D., Sullivan, D., Reimers, M. & Waters, J. (2020). Separation of hemodynamic signals from GCaMP fluorescence measured with wide-field imaging. *Journal of Neurophysiology*, 123(1), 356–366. doi:10.1152/jn.00304.2019
- [9] Ma, Y., Shaik, M.A., Kim, S.H., Kozberg, M.G., Thibodeaux, D.N., Zhao, H.T., Yu, H. & Hillman, E.M.C. (2016). Wide-field optical mapping of neural activity and brain haemodynamics: considerations and novel approaches. *Philosophical Transactions of the Royal Society B*, 371(1705), 20150360. doi:10.1098/rstb.2015.0360
- [10] Ren, C. & Komiyama, T. (2021). Characterizing cortex-wide dynamics with wide-field calcium imaging. *Journal of Neuroscience*, 41(19), 4160–4168. doi:10.1523/JNEUROSCI.3003-20.2021 — <https://www.jneurosci.org/content/41/19/4160>
- [11] Ren, C. & Komiyama, T. (2021). Wide-field calcium imaging of cortex-wide activity in awake, head-fixed mice. *STAR Protocols*, 2(4), 100973. doi:10.1016/j.xpro.2021.100973 — <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8608657/>
- [12] *wfield*: tools to process widefield data (open-source software; motion correction, penalised matrix decomposition, hemodynamic correction, LocaNMF). <https://github.com/churchlandlab/wfield>
- [13] *WidefieldImager*: pre-processing pipeline for widefield data (open-source MATLAB software; localised SVD dimensionality reduction and hemodynamic correction). <https://github.com/churchlandlab/WidefieldImager>
- [14] Neuroscience Cloud Analysis As a Service: an open-source platform for scalable, reproducible data analysis. *Neuron* (2022). [https://www.cell.com/neuron/fulltext/S0896-6273\(22\)00587-6](https://www.cell.com/neuron/fulltext/S0896-6273(22)00587-6)
- [15] Xiao, D., Forsy, B.J., Vanni, M.P. & Murphy, T.H. (2021). MesoNet allows automated scaling and segmentation of mouse mesoscale cortical maps using machine learning. *Nature Communications*, 12(1), 5992. doi:10.1038/s41467-021-26255-2 — <https://www.nature.com/articles/s41467-021-26255-2>
- [16] Weiser, S.C., Mullen, B.R., Ascencio, D. & Ackman, J.B. (2023). Data-driven segmentation of cortical calcium dynamics. *PLOS Computational Biology*, 19(5), e1011085. doi:10.1371/journal.pcbi.1011085 — <https://pmc.ncbi.nlm.nih.gov/articles/PMC10174627/>
- [17] Linden, N.J., Tabuena, D.R., Steinmetz, N.A., Moody, W.J., Brunton, S.L. & Brunton, B.W. (2021). Go with the FLOW: visualizing spatiotemporal dynamics in optical widefield calcium imaging. *Journal of the Royal Society Interface*, 18(181), 20210523. doi:10.1098/rsif.2021.0523
- [18] Wang, Q., Ding, S.-L., Li, Y., et al. (2020). The Allen Mouse Brain Common Coordinate Framework: a 3D reference atlas. *Cell*, 181(4), 936–953.e20. doi:10.1016/j.cell.2020.04.007
- [19] Wexselblatt, J.B., Flister, E.D., Piscopo, D.M. & Niell, C.M. (2016). Large-scale imaging of cortical dynamics during sensory perception and behavior. *Journal of Neurophysiology*, 115(6), 2852–2866. doi:10.1152/jn.01056.2015
- [20] Nietz, A.K., Popa, L.S., Streng, M.L., Carter, R.E., Kodandaramaiah, S.B. & Ebner, T.J. (2022). Wide-field calcium imaging of neuronal network dynamics in vivo. *Biology*, 11(11), 1601. doi:10.3390/biology11111601 — <https://pmc.ncbi.nlm.nih.gov/articles/PMC9687960/>
- [21] An open source statistical and data processing toolbox for wide-field optical imaging in mice (preprint). bioRxiv, 2021. <https://www.biorxiv.org/content/10.1101/2021.04.07.438885>
- [22] Rossi, L.F., Kullmann, D.M. & Wykes, R.C. (2018). The enlightened brain: novel imaging methods focus on epileptic networks at multiple scales. *Frontiers in Cellular Neuroscience*, 12, 82. doi:10.3389/fncel.2018.00082
- [23] Musall, S., Kaufman, M.T., Juavinett, A.L., Gluf, S. & Churchland, A.K. (2019). Single-trial neural dynamics are dominated by richly varied movements. *Nature Neuroscience*, 22(10), 1677–1686. doi:10.1038/s41593-019-0502-4
- [24] Shore, A.N., Colombo, S., Tobin, W.F., et al. & Weston, M.C. (2020). Reduced GABAergic neuron excitability, altered synaptic connectivity, and seizures in a KCNT1 gain-of-function mouse model of childhood epilepsy. *Cell Reports*, 33, 108303. doi:10.1016/j.celrep.2020.108303
- [25] Mesoscale Imaging of Neural Dynamics in Epilepsy. In: Niemeyer, J.E., Ma, H. & Schwartz, T.H. (eds), *Awake Behaving Mesoscopic Brain Imaging*, Neuromethods vol. 214. Humana, New York (2025). doi:10.1007/978-1-0716-4120-0_11
- [26] Luo, P., Yang, F., Li, J., Niemeyer, J.E., et al. & Schwartz, T.H. (2023). Excitatory-inhibitory mismatch shapes node recruitment in an epileptic network. *Epilepsia*, 64, 1939. doi:10.1111/epi.17638

- [27] Waters, J. (2020). Sources of widefield fluorescence from the brain. *eLife*, 9, e59841. doi:10.7554/eLife.59841
- [28] Decoding cortical brain states from widefield calcium imaging data using visibility graph (GCaMP6f cortical recordings; comparison of feature sets and classifiers). <https://www.researchgate.net/publication/325634015>
- [29] Wang, L.-H., Zhang, Z.-N., Xie, C.-X., et al. (2025). A novel real-time threshold algorithm for closed-loop epilepsy detection and stimulation system. *Sensors*, 25(1), 33. doi:10.3390/s25010033 — <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11723055/>
- [30] Acharya, U.R., Oh, S.L., Hagiwara, Y., Tan, J.H. & Adeli, H. (2018). Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals. *Computers in Biology and Medicine*, 100, 270–278. doi:10.1016/j.combiomed.2017.09.017
- [31] Asif, U., Roy, S., Tang, J. & Harrer, S. (2019). SeizureNet: multi-spectral deep feature learning for seizure type classification. arXiv:1903.03232. <https://arxiv.org/abs/1903.03232>
- [32] End-to-end deep learning approach to mouse behavior classification from cortex-wide calcium imaging. *PLOS Computational Biology* (2023), 19, e1011074. doi:10.1371/journal.pcbi.1011074
- [33] Deep learning in intracranial EEG for seizure detection: advances, challenges, and clinical applications. *Frontiers in Neuroscience* (2025), 19, 1677898. doi:10.3389/fnins.2025.1677898
- [34] A deep learning approach for automatic seizure detection in children with epilepsy. *Frontiers in Computational Neuroscience* (2021), 15, 650050. doi:10.3389/fncom.2021.650050
- [35] Epilepsy detection based on spatiotemporal feature interaction fusion of EEG signals (graph attention plus transformer). *Frontiers in Neurology* (2025), 16, 1478718. doi:10.3389/fneur.2025.1478718
- [36] Expert-level intracranial electroencephalogram ictal pattern detection by a deep learning neural network. *Frontiers in Neurology* (2021). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8126697/>